

ISKANDAR KHOLMANOV

DATA SCIENTIST/ML ENGINEER

iskandar.kholmanov@gmail.com

512-825-4935

Austin, TX

[LinkedIn](#)

[Github](#)

Green card

U.S. Work Authorization – No visa sponsorship required (now or in the future)

SUMMARY

Data Scientist/ML Engineer with a PhD in Physics and 8+ years solving complex, data-intensive problems across research and industry. Hands-on experience in Python, SQL, building and deploying cloud-native ML systems with API-based architectures and scalable data pipelines, and end-to-end GenAI systems, including LLM fine-tuning, RAG pipelines, and production deployments. Proven ability to design and deliver reliable, scalable ML systems for real-world applications and production deployment. Co-author of 40+ scientific publications and a U.S. patent.

SKILLS

- ❖ **Programming & MLOps:** Python, SQL, Git, Docker, CI/CD (GitHub Actions), Databricks (Spark), AWS (EC2, S3, Lambda, API Gateway), Serverless ML, model serving, MLflow, Airflow, Evidently, cloud-native deployment;
- ❖ **GenAI & LLMs:** LLM fine-tuning (LoRA/PEFT), RAG pipelines, LangChain, LlamaIndex, Hugging Face, VectorDB (FAISS, ChromaDB), semantic search;
- ❖ **ML & DL:** PyTorch, TensorFlow, scikit-learn, classical ML, deep learning, transformer architectures;
- ❖ **Statistical Analysis & Experimentation:** EDA, A/B testing, time series analysis, data visualization, statmodels;
- ❖ **Leadership & Operations:** Cross-functional collaboration, data-driven decision-making, end-to-end ownership, operational optimization

EXPERIENCE

- ❖ **Data Scientist / Machine Learning Engineer** | TripleTen | 2024–Present
 - Designed, built, and deployed ML and GenAI systems in education and production environments;
 - Developed RAG pipelines, LLM fine-tuning workflows, and scalable ML models;
 - Led end-to-end ML lifecycle: data, modeling, deployment, monitoring;
- ❖ **Business owner** | 2017-2024
 - Provided decisive COVID-19 crisis leadership, preserving margins through forecasting and cost control.;
 - Led strategic planning to drive revenue growth and profitability across key business units.
- ❖ **Research Scientist**, The University of Texas at Austin | 2010 – 2017
 - Led data-driven research resulting in 40+ papers, a U.S. patent, and industry partnerships. | [Github](#)

TECH PROJECTS

- ❖ **End-to-end RAG system with decoupled Hugging Face space deployment.** Developed a production-ready, scalable **LangChain** RAG system using a **Wikidata SPARQL** ML knowledge, **sentence-transformers /all-MiniLM-L6-v2** embeddings, **ChromaDB** for vector retrieval, and **FLAN-T5-Large** for offline generation; deployed a **Dockerized FastAPI backend** and **Gradio frontend** on separate **Hugging Face Spaces** with automated (Github Actions) **CI/CD** pipelines and no external API dependencies. | [Github](#) | [Hugging Face](#)
- ❖ Built an **end-to-end LLM fine-tuning** pipeline (**Mistral-7B + LoRA/PEFT**) using **Google Colab (T4 GPU)** for efficient training and experimentation. Created instruction-tuning datasets (system/user/assistant format) using Amazon reviews, synthetic Q&A, and product metadata. Deployed a **4-bit quantized inference** pipeline (**QLoRA**) with **FastAPI**, exposed via **ngrok for real-time chat**. Analyzed model behavior and identified dataset-scale and instruction-alignment limits affecting LoRA performance. | [Github](#)
- ❖ **Stock Price Forecasting.** Built a production-grade time-series ML system using **PyTorch LSTM**, orchestrated with **Dockerized Airflow**, tracked via **MLflow**, monitored with **Evidently**, and **deployed on Streamlit Cloud** for live inference; ingested Yahoo Finance data; benchmarked against **SARIMA**, **Random Forest**, **XGBoost**, and **LSTM-Keras** models. The PyTorch LSTM achieved **1.64% MAPE** on unseen data, outperforming all baselines | [Github](#) | [Streamlit App](#)

EDUCATION

- **TripleTen Bootcamp** | Data Science / Machine Learning Certificate | 2024;
- **PhD in Physics**, Academy of Science of Uzbekistan;